
Amazon Comprehend: NLP Powered by AWS

Amazon Comprehend is a **Natural Language Processing (NLP)** service from AWS that uses **machine learning** to uncover **insights and relationships in unstructured text** — such as customer reviews, support tickets, social media, and documents.

What Can Amazon Comprehend Do?

1 Entity Recognition

- Detects entities such as:
 -  Person names
 -  Locations
 -  Organizations
 -  Dates, quantities, currencies
 - Example: “Amazon is headquartered in Seattle.” → Entities: Amazon (ORG), Seattle (LOC)
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2 Sentiment Analysis

- Determines overall sentiment of text:
 -  Positive

- 😐 Neutral
 - 😞 Negative
 - 😐 Mixed
- Supports **multiple languages** (English, Spanish, German, etc.)
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3 Key Phrase Extraction

- Extracts **important concepts or terms**
 - Example: From “AWS offers scalable cloud services,” it extracts:
 - “scalable cloud services”
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4 Language Detection

- Identifies the **dominant language** in a text snippet
 - Useful for multilingual apps, auto-routing, localization
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5 Syntax Analysis

- Breaks text into:
 - Part of speech (noun, verb, etc.)
 - Tokenization (word-level analysis)

6 PII Detection (Personally Identifiable Information)

- Finds and redacts sensitive information:
 - Names, phone numbers, credit card numbers, SSNs, emails, etc.
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7 Custom Classification & Entity Recognition

- Train your own models for:
 - Detecting **domain-specific labels** (e.g., Support vs. Complaint)
 - Custom **entity types** (e.g., Product IDs, Ticket Numbers)
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Use Cases

Scenario	Use Case
 E-commerce	Analyze product reviews or customer service chats
 Call Centers	Classify and prioritize support tickets
 Business Intelligence	Extract insights from survey responses
 Compliance	Detect and redact PII in stored documents

Sample CLI/API Usage

CLI Example:

```
aws comprehend detect-sentiment \
```

```
--text "I love using AWS services!" \  
--language-code en
```

Output:

```
{  
  "Sentiment": "POSITIVE",  
  "SentimentScore": {  
    "Positive": 0.98,  
    "Negative": 0.01,  
    "Neutral": 0.01,  
    "Mixed": 0.00  
  }  
}
```

Security & Compliance

- Data encrypted **at rest and in transit**
 - No data stored unless explicitly enabled
 - Supports **HIPAA** and other compliance frameworks
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Pricing Overview

- Pay-per-use: Billed per number of characters analyzed
- Separate pricing for synchronous vs. batch jobs
- Free tier: 50,000 characters/month for 12 months

 [Official Pricing Page](#)
